

BINARY ARITHMETIC PRACTICE SHEET

NIMCET TARGET PAPER • EASY DIFFICULTY LEVEL

ResourceOS

1. What is the sum of the binary numbers 1010 and 0101 ?

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|----------|----------|
| (A) 1111 | (B) 1001 |
| (C) 1100 | (D) 1010 |

2. Perform binary addition: $1101_2 + 1011_2$.

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|-----------|-----------|
| (A) 11000 | (B) 10100 |
| (C) 11100 | (D) 10011 |

3. What is the result of binary subtraction: $1100_2 - 0101_2$?

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|----------|----------|
| (A) 0111 | (B) 0110 |
| (C) 1001 | (D) 0101 |

4. Multiply the binary numbers 101_2 and 11_2 .

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|----------|----------|
| (A) 1111 | (B) 1101 |
| (C) 1001 | (D) 1011 |

5. What is the quotient when 1100_2 is divided by 100_2 ?

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|---------|---------|
| (A) 11 | (B) 10 |
| (C) 101 | (D) 100 |

6. What is the 1's complement of the binary number 10110010 ?

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|--------------|--------------|
| (A) 01001101 | (B) 01001110 |
| (C) 10110011 | (D) 01010010 |

7. What is the 2's complement of the binary number 10101000 ?

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|--------------|--------------|
| (A) 01011000 | (B) 01010111 |
| (C) 01010100 | (D) 10101001 |

8. Find the 8-bit 2's complement representation of decimal -5 .

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|--------------|--------------|
| (A) 11111011 | (B) 11111010 |
| (C) 10000101 | (D) 11111101 |

9. In an 8-bit 2's complement system, what decimal value is represented by 11111100 ?

- | | |
|------------|------------|
| (A) -4 | (B) -3 |
| (C) -252 | (D) $+252$ |

10. What is the binary sum of $1+1+1$ in basic binary addition rules?

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|------------------------|------------------------|
| (A) Sum = 1, Carry = 1 | (B) Sum = 0, Carry = 1 |
| (C) Sum = 1, Carry = 0 | (D) Sum = 3, Carry = 0 |

11. Calculate binary subtraction: $1000_2 - 0001_2$.

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|----------|----------|
| (A) 0111 | (B) 0110 |
| (C) 1001 | (D) 0011 |

12. Multiply binary 110_2 by 101_2 .

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|-----------|-----------|
| (A) 11110 | (B) 11010 |
| (C) 10110 | (D) 11000 |

13. Divide 10100_2 by 100_2 .

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|---------|---------|
| (A) 101 | (B) 110 |
| (C) 11 | (D) 100 |

14. What is the 2's complement of 00000000 in an 8-bit representation?

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|--------------|--------------|
| (A) 00000000 | (B) 11111111 |
| (C) 10000000 | (D) 00000001 |

15. In 2's complement arithmetic, subtraction $A - B$ is performed as:

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|---|---|
| (A) $A + (2\text{'s complement of } B)$ | (B) $A + (1\text{'s complement of } B)$ |
| (C) $(2\text{'s complement of } A) - B$ | (D) $(1\text{'s complement of } A) + B$ |

16. Find the value of $1011_2 + 1101_2 + 0001_2$.

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|-----------|-----------|
| (A) 11001 | (B) 11000 |
| (C) 10111 | (D) 11101 |

17. Convert decimal 15 to binary and find its 1's complement in 8-bit format.

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|--------------|--------------|
| (A) 11110000 | (B) 00001111 |
| (C) 11110001 | (D) 11101111 |

18. What is the 8-bit 2's complement representation of decimal -1 ?

- | | |
|--------------|--------------|
| (A) 11111111 | (B) 10000001 |
| (C) 11111110 | (D) 01111111 |

19. Calculate: $11100_2 \div 100_2$.

- (A) 111
(C) 101

- (B) 110
(D) 100

20. What is the product of 111_2 and 101_2 ?

- (A) 100011
(C) 101011

- (B) 100101
(D) 011111

21. In binary subtraction using 2's complement, if there is a carry out from the MSB, the result is:

- (A) Positive, and carry is discarded
(C) Zero

- (B) Negative, and carry is added
(D) Invalid / Overflow

22. What is the range of signed decimal integers that can be represented using 4-bit 2's complement?

- (A) -8 to +7
(C) -8 to +8

- (B) -7 to +7
(D) 0 to +15

23. Perform 8-bit 2's complement addition: $00000110 (+6) + 11111010 (-6)$.

- (A) 00000000 (with carry discarded)
(C) 00001100

- (B) 11111111
(D) 11110000

24. Shift left by 1 position ($1011_2 \ll 1$) is equivalent to:

- (A) Multiplying by 2
(C) Adding 2

- (B) Dividing by 2
(D) Subtracting 2

25. What is the value of $10100_2 - 1010_2$ in decimal?

- (A) 10
(C) 8

- (B) 12
(D) 14

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)